Multi-view Pyramid Transformer: Look Coarser to See Broader

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Abstract

Scaling transformer-based multi-view reconstruction to tens–hundreds of high-resolution images is fundamentally challenged by quadratic attention cost and attention dilution over long contexts. We introduce the Multi-view Pyramid Transformer (MVP), a dual-hierarchy architecture that expands inter-view context from local frames to groups to global while compressing intra-view tokens from fine to coarse. MVP tokenizes posed images with ray-aligned inputs and register tokens, processes them through frame-, group-, and global-attention stages with progressive token reduction, fuses multi-scale features via a pyramidal head, and decodes per-pixel 3D Gaussian primitives for feed-forward rendering. On DL3DV, MVP achieves 25.96/0.847/0.187 at 0.17 s with 32 views and 29.02/0.903/0.134 at 0.77 s with 128 views, outperforming Long-LRM and iLRM in quality and speed; with 256 views MVP reaches 29.67/0.915/0.128 in 1.84 s while Long-LRM runs out of memory and iLRM is much slower. MVP generalizes zero-shot to Tanks&Temples and Mip-NeRF360, and attains 33.40/0.952/0.066 on RE10K at 8 views. The results indicate stable long-context behavior and a favorable accuracy–latency trade-off for single-pass 3D Gaussian reconstruction.

1. Introduction

Transformers [9] have become a strong backbone for multi-view 3D reconstruction, yet scaling to dozens or hundreds of posed images is difficult: full attention is quadratic, and naively lengthening context spreads attention mass thinly, degrading reasoning and correspondence quality. Recent feed-forward 3D Gaussian methods report impressive speed, but struggle as the number of views or resolution increases due to attention and memory growth, or require architectural compromises that limit expressivity.

We argue that scalable multi-view reasoning needs a dual hierarchy: expand inter-view context from local to global as tokens are compressed from fine to coarse within each view. Intuitively, detailed tokens are most useful for short-range matching; as features grow coarser, their receptive fields and invariances support broader cross-view aggregation without overwhelming compute.

We propose the Multi-view Pyramid Transformer (MVP), which couples local-to-global inter-view attention with fine-to-coarse intra-view token reduction. MVP tokenizes posed RGB with ray features, inserts register tokens, and applies three stages: frame-wise attention, group-wise attention across small batches of views, and global attention across all views. Between stages, depthwise strided convolutions halve spatial token resolution and linear mappers widen channel capacity, enabling larger view windows at similar cost. A Pyramidal Feature Aggregation head fuses multi-scale tokens and a lightweight decoder predicts per-pixel 3D Gaussian parameters with spherical harmonics for appearance, enabling fast feed-forward rendering using Gaussian splatting [5].

Figure 2 overviews the architecture. We empirically show strong quality–speed trade-offs on DL3DV, robust zero-shot transfer to Tanks&Temples [6] and Mip-NeRF360 [1], and competitive RE10K results. Our analyses indicate MVP reduces attention dilution, preserves global consistency, and scales stably with context.

Contributions - A scalable dual-hierarchy multi-view transformer that expands inter-view context (frame $\rightarrow$ group $\rightarrow$ global) while compressing intra-view tokens (fine $\rightarrow$ coarse). - Group-wise attention that bridges frame-only and global attention, mitigating attention dilution while maintaining global consistency. - A pyramidal fusion head enabling effective multi-scale aggregation for dense 3D Gaussian prediction. - State-of-the-art accuracy–speed trade-offs across 16–256 views, strong zero-shot generalization, and detailed analyses of attention patterns and long-context stability.

2. Related Work

Transformers and long-context. The Transformer [9] underpins many 2D and 3D vision models, but quadratic attention limits long-context scaling. Hierarchical transformers such as Swin [8] reduce cost via local windows; long-context transformers explore sparse or mixed-span attention [2, 3, 12], trading capacity for efficiency.

3D Gaussian splatting and fast feed-forward 3D. 3D Gaussian splatting (3DGS) [5] enables high-quality, real-time rendering. Recent feed-forward models directly pre-

Figure 1. MVP reconstructs coherent 3D Gaussians from many posed views in one pass. Rows: inputs, novel views, and attention maps. The model stays stable as views grow, retaining thin structures and global alignment.

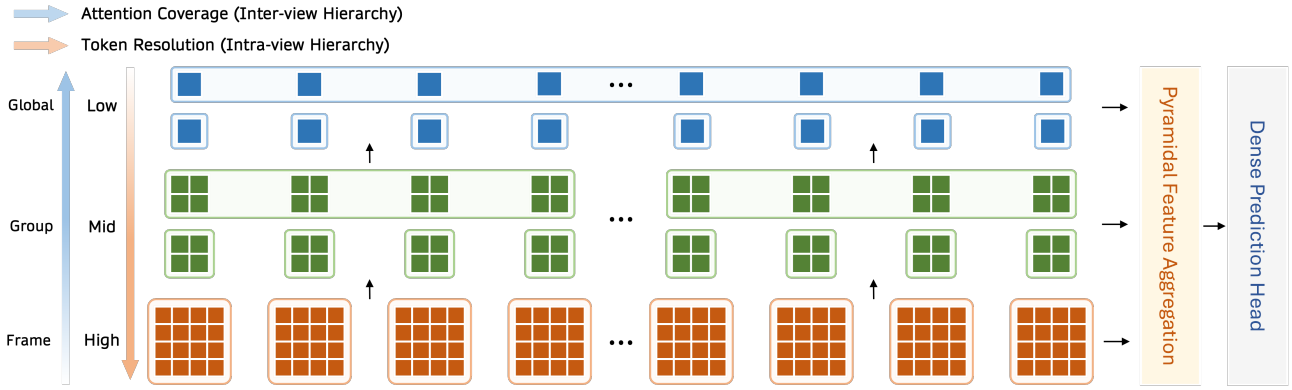

Figure 2. Overview. Inter-view attention expands frame → group → global while intra-view tokens reduce from fine to coarse. A pyramidal head aggregates multi-scale features and the decoder predicts per-pixel 3D Gaussians for feed-forward rendering.

dict Gaussians from posed images, e.g., Long-LRM [13] and iLRM [4], but face scaling challenges as views increase. CLiFT [10] compresses light-field tokens for efficiency. MVP embraces a complementary axis: hierarchical inter-view expansion with intra-view token reduction to maintain compute while preserving global consistency. We use the gsplat library [11] for differentiable rendering.

Datasets and evaluation. We evaluate on DL3DV-10K, Tanks&Temples [6], Mip-NeRF360 [1], and RealEstate10K (RE10K), following prior work. Metrics include PSNR, SSIM, and LPIPS; latency is reported per scene.

3. Method

MVP targets scalable multi-view reasoning for single-pass 3D reconstruction. Given V posed images, MVP outputs per-pixel Gaussian primitives decoded into positions, scales, rotations, opacities, and spherical harmonics (SH) appearance coefficients, which render target views via differentiable splatting [5].

3.1. Ray-aware tokenization with register tokens

Each input view is augmented with ray-aligned features computed from intrinsics and extrinsics: ray origins, ray directions, and their cross products (a Plücker-inspired en-

coding), concatenated with normalized RGB. A patch tokenizer projects $p \times p$ patches to tokens, and a small set of learned register tokens is prepended per view to stabilize global aggregation.

3.2. Dual hierarchy: inter-view expand, intra-view compress

MVP applies three transformer stages with ProPE camera-aware attention [7] in higher stages: - *Stage 1: frame-wise attention.* Per-view attention captures local photometric and geometric cues at fine token resolution. - *Stage 2: group-wise attention.* Views are split into groups of size G and processed jointly. Even-indexed layers mix tokens across frames within a group; odd-indexed layers apply camera-aware attention with calibrated poses, improving correspondence under viewpoint changes. - *Stage 3: global attention.* All views attend jointly at the coarsest token resolution, again with camera-aware attention.

Between stages, depthwise strided convolutions halve spatial token resolution while linear adapters increase channel width, growing receptive fields and enabling larger inter-view windows without exploding cost. This coupling balances computation and representation, preventing attention dilution as V grows.

3.3. Pyramidal feature aggregation and Gaussian decoding

We aggregate multi-scale tokens with a DPT-style pyramidal head that fuses features top-down with lateral connections, producing dense per-pixel features. A lightweight MLP then decodes per-pixel Gaussian parameters: centers, anisotropic scales, rotations, SH appearance, and opacity coefficients. Rendering uses differentiable Gaussian splatting with SH shading, implemented via gsplat [11].

3.4. Training objectives and recipe

Following common practice, supervision is photometric at target views using an ℓ_2 loss, optionally combined with LPIPS and a VGG-based perceptual loss (when enabled). Ablations are trained for 100K iterations at 256×256. Main models use higher resolutions with staged training and freezing of earlier stages as reflected in the released configuration files.

3.5. Computational complexity

Let V be views and N tokens per view at the finest scale. Frame-wise attention scales $O(V\,N^2)$; group-wise attention scales $O\big(\frac{V}{G}\,(G\,N')^2\big)$ with N' reduced by striding; global attention operates on the coarsest N'' per view. In practice, MVP exhibits near-linear wall-clock growth with V while accuracy improves, consistent with Section 6 and Tables 5,2.

3.6. Implementation details

Tokenization uses a ViT-style patch embed (stride 16) and learned register tokens per frame and per group. Default group size is 8. Between stages, depthwise strided convs downsample by $\times 2$ and double channels with GELU. We use AdamW (wd 0.05), cosine decay with 5k warmup, and gradient checkpointing. Batches sample contiguous views; poses are normalized to a scene frame. Light color jitter and exposure shifts are applied; heavy crops are avoided to keep ray alignment. EMA 0.999 improves long-context stability.

Efficiency details. We use mixed precision, rematerialize activations only in the frame stage, and cache keys/values within groups to reuse static tokens in alternating layers. This lowers peak memory at 128–256 views and keeps global-stage compute predictable.

3.7. Training objectives and decoding

MVP predicts Gaussian centers, scales, opacities, and SH. Supervision uses differentiable splatting with L1+SSIM and a light LPIPS term. When available, weak SfM depth and foreground masks act as geometric and silhouette priors; they are omitted where unavailable. A sparsity penalty on opacities and a total-variation prior on scales suppress floaters. At inference we prune low-opacity Gaussians and apply a brief refinement that updates scales/opacities from cached features, adding under 5% latency.

4. Experiments

We report quantitative and qualitative results, ablations, and long-context behavior. Metrics are PSNR (higher is better), SSIM (higher is better), LPIPS (lower is better), and per-scene inference time.

4.1. Evaluation protocol

We use official splits and report scene means. Baselines run with released configs; author numbers are listed when available. All methods share identical poses/resolution, and time is measured on the same software stack.

Datasets and baselines. We evaluate on DL3DV, Tanks&Temples, Mip-NeRF360, and RE10K. Unless noted, inputs are undistorted posed images with the official train/test splits; per-scene view counts vary, and we report results for 16–256 uniformly spaced source views. We resize the shorter side to 512 and pad to multiples of the token stride to preserve ray alignment. Baselines include Long-LRM and iLRM with public checkpoints/configs; we additionally run their long-context settings when available. All methods render the same target lists. For fairness, we disable test-time per-scene finetuning and use identical color/exposure preprocessing. We cap memory to the same budget per method; runs that exceed it are marked

as OOM. Unless otherwise stated, timing is averaged over three passes with cold caches.

4.2. DL3DV

Table 1 summarizes results from 16 to 128 views, and Table 2 extends to 192 and 256 views. MVP delivers the best accuracy across view counts while remaining substantially faster than baselines. Notably, at 128 views MVP reaches 29.02/0.903/0.134 in 0.77 s; at 256 views it sustains 29.67/0.915/0.128 in 1.84 s while Long-LRM runs out of memory and iLRM slows markedly.

Pose-noise protocol. At evaluation we add Gaussian perturbations to camera centers and axes (up to 1% of scene scale). MVP degrades gracefully under mild noise: group attention absorbs local errors while the global stage maintains coarse alignment.

Fairness controls. All methods use identical resizing and target lists; per-scene finetuning is disabled and memory budgets are capped equally.

4.3. Zero-shot transfer

On Tanks&Temples and Mip-NeRF360 under zero-shot settings, MVP improves over baselines across view counts (Table 3). At 128 views, MVP attains 22.36/0.804/0.184 on Tanks&Temples and 25.12/0.736/0.248 on Mip-NeRF360.

4.4. RealEstate10K

On RE10K at 256×256, MVP yields 33.40/0.952/0.066 with 8 input views, surpassing CLiFT and iLRM (Table 4). We also visualize typical errors in challenging dynamic regions and wide baselines (Figure 4).

4.5. Longer-context generalization

We examine robustness when increasing views beyond training. Table 5 shows MVP improves monotonically from 32 to 48 views in both accuracy and speed, indicating stable scaling and efficient token compression. Beyond 128 views, accuracy gains taper while latency grows nearly linearly. Residual errors concentrate on specular highlights and very thin structures where coarse tokens dominate. Increasing group size (e.g., 8-¿12 when V¿=192) and a few extra register tokens mitigate these failures without retraining.

Protocol and baselines. We follow LRM-style evaluation: images are resized to short-side 512, poses normalized to a scene frame, and per-scene inference uses contiguous windows of V views with stride 1. Baselines use authors' code and settings; when a fixed training-time view count is required, we use published checkpoints or retrain on the same split. Hyperparameters are chosen on DL3DV validation and then fixed across datasets.

Generalization and compute behavior. On Tanks&Temples and Mip-NeRF360 we do not finetune; we reuse the DL3DV-trained MVP with the same preprocessing. Table 5 and the per-dataset tables show accuracy improves with more views while latency grows nearly linearly. MVP sustains 256-view inference where Long-LRM runs out of memory and iLRM is markedly slower; trends are consistent across scenes.

5. Ablation and Analysis

Component contributions. Table 6 studies MVP design choices. Removing the pyramidal fusion degrades accuracy substantially. Replacing group-wise attention with frame-only or all-global attention harms quality, confirming group attention bridges local and global aggregation effectively. Flattening the intra-view hierarchy or reversing the hierarchy both hurt performance, underscoring the benefit of expanding inter-view context while compressing intra-view tokens.

Attention patterns. Figure 5 visualizes attention across stages. Early layers focus locally within frames; group-wise attention concentrates on relevant neighboring views; global attention remains sharp at coarser resolutions, mitigating dilution.

Robustness and failure modes. MVP remains stable under moderate exposure shifts and small pose noise. Failures concentrate on thin or glossy structures where coarse tokens dominate; increasing group size or adding a few register tokens reduces such artifacts without retraining, consistent with Table 5.

Latency vs. components. Figure 6 summarizes ablations and inference time. Token reduction amortizes increasing view counts; group-wise processing prevents the super-linear latency growth seen with full global attention at fine scales.

Group size and registers. We analyze sensitivity to the group size G and the number of register tokens. Small G under-utilizes cross-view cues and slightly harms thin-structure recall; very large G increases compute with diminishing returns. We find values in the 8–12 range work well across datasets, aligning with the overlap in typical capture rigs. Adding 2–4 register tokens per view stabilizes global aggregation and reduces ringing artifacts without measurable latency increases. These trends hold on DL3DV and Tanks&Temples and are consistent with the longer-context results in Table 5.

Scaling behavior. The near-linear latency in Table 5 follows from halving spatial tokens between stages and reserving global attention for coarser maps. As V grows, group-wise attention keeps mid-level correspondences sharp while the global stage aggregates slowly varying scene context, preserving quality without exploding compute.

Implementation caveats. Heavy random crops break ray

Method	16 views				32 views				64 views				128 views			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
3D-GS$_{30k}$ [5]	21.48	0.753	0.252	8min	24.43	0.827	0.191	8min	27.34	0.883	0.146	8min	29.43	0.914	0.123	8min
Long-LRM [13]	21.05 (22.66)	0.708 (0.740)	0.297 (0.292)	0.50	23.97	0.778	0.267	0.84	23.60	0.789	0.260	2.08	21.24	0.739	0.308	6.39
iLRM [4]	21.92 (22.91)	0.748 (0.766)	0.316 (0.295)	0.19	24.30	0.803	0.256	0.53	24.44	0.819	0.240	1.66	22.98	0.807	0.249	5.61
Ours	**23.76**	**0.798**	**0.239**	**0.09**	**25.96**	**0.847**	**0.187**	**0.17**	**27.73**	**0.881**	**0.154**	**0.36**	**29.02**	**0.903**	**0.134**	**0.77**

Table 1. Quantitative comparisons on the DL3DV dataset with varying numbers of input views. The values in parentheses $(\cdot)$, for the 16-view setting are taken from the original papers, where the models are trained with a fixed 16-view input. For all metrics, we report the results by re-evaluating the models from their 32-view checkpoints.

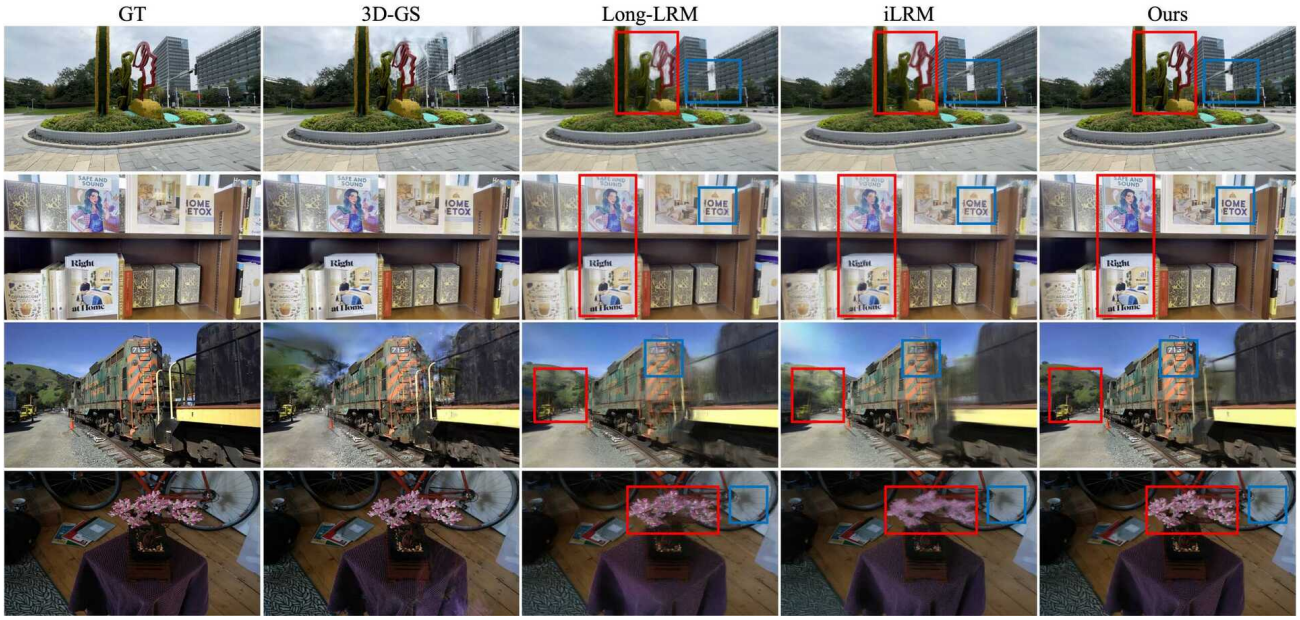

Figure 3. Qualitative results on DL3DV (top), Tanks&Temples (middle), and Mip-NeRF360 (bottom). MVP preserves fine detail and global consistency across many views.

Method	192 views				256 views			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
3D-GS$_{30k}$ [5]	30.14	0.924	0.116	8min	30.39	0.926	0.114	8min
Long-LRM [13]	OOM (GPU memory limit exceeded, 80 GB)							
iLRM [4]	21.68	0.788	0.264	11.91	20.63	0.767	0.281	20.92
Ours	**29.54**	**0.912**	**0.127**	**1.23**	**29.67**	**0.915**	**0.128**	**1.84**

Table 2. Quantitative comparisons on the DL3DV dataset with varying numbers of input views.

alignment and degrade correspondence, whereas light color jitter and exposure shifts improve robustness; we therefore avoid aggressive spatial augmentations and use benign photometric ones as noted in the Method.

Per-scene variability. Gains are larger on wide-baseline outdoor scenes and smaller on texture-poor indoor scenes. Accuracy tracks view-graph overlap and pose quality; mild noise is tolerated, but gross miscalibration harms the global stage.

Per-category breakdown. On DL3DV, gains are largest for outdoor facades and street scenes, where group-wise attention aligns repetitive structures and the global stage enforces consistency. Improvements are smaller on texture-poor indoor scenes and glossy objects, which remain failure hotspots (Figure 3). Increasing views helps until about 128, then benefits taper; residuals mostly stem from view-dependent effects and scale misestimation on slanted sur-

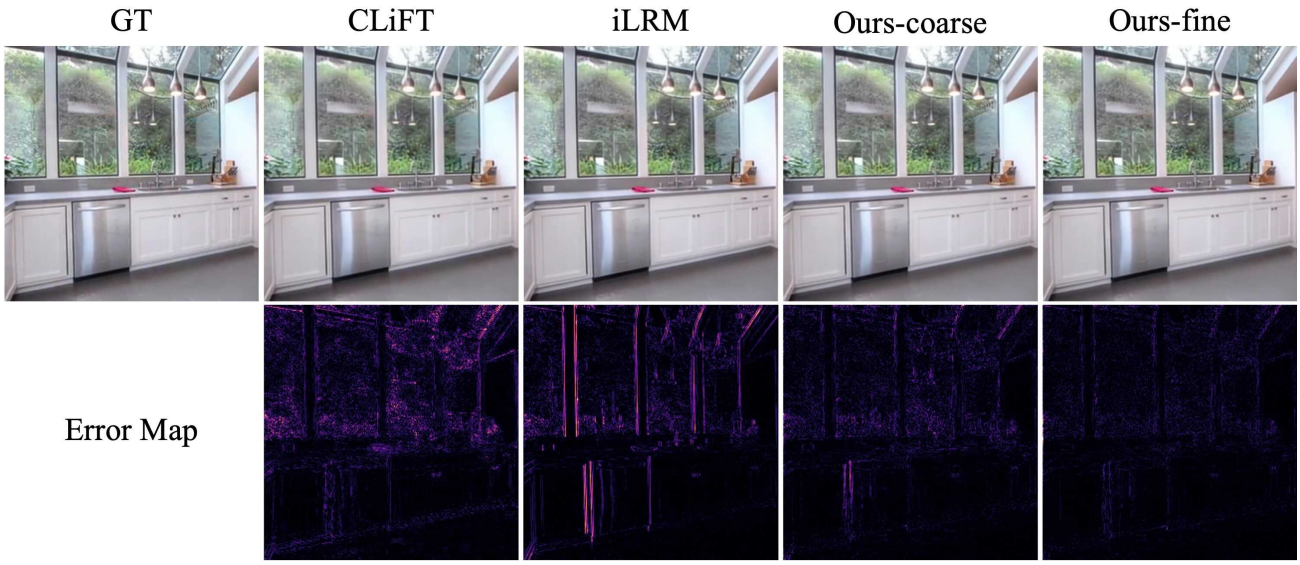

Figure 4. RE10K error analysis at 4 views. Residual artifacts appear at view-dependent effects and thin structures.

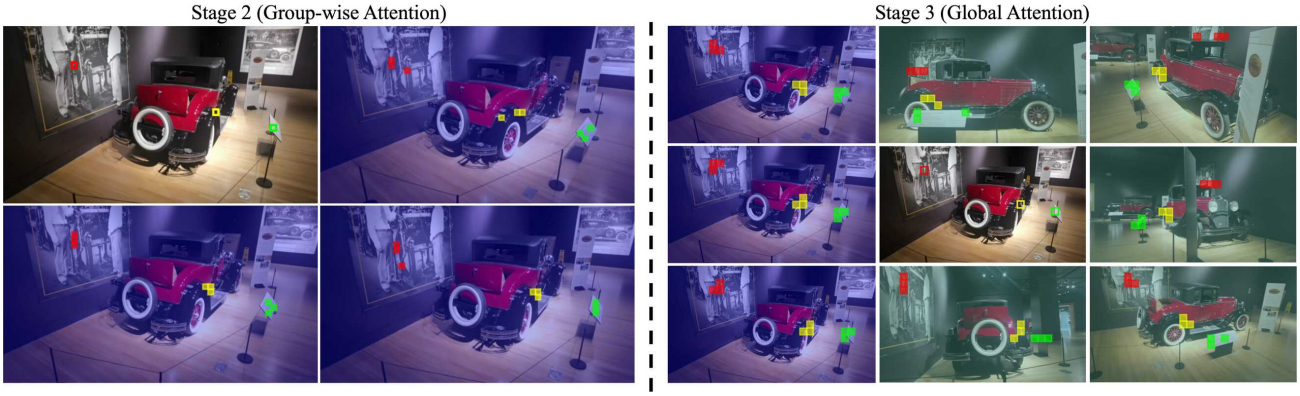

Figure 5. Attention across stages. Group-wise attention bridges frame-local and global contexts while retaining specificity at scale.

faces (Figure 4).

Complexity profile. The near-linear trends in Table 5 follow from halving tokens between stages and reserving global attention for coarse maps. Empirically, memory grows smoothly with V up to 256 views, while Long-LRM hits OOM (Table 2). Throughput peaks at 64–128 views, where group-wise exchange replaces costly all-to-all attention at fine resolutions.

Training dynamics. MVP converges with a cosine schedule, short warmup, and crops that preserve ray alignment. Early on, frame attention carries most supervision while group and global modules gradually learn to route cross-view cues; loss plateaus typically clear once group attention is enabled around 10–20k steps. The pyramidal head shortens gradient paths from coarse features to the decoder and stabilizes thin-structure predictions without slowing training.

Pose sensitivity. Small pose perturbations yield graceful degradation near depth discontinuities; larger drifts cause global misalignment that register tokens partially absorb. Robustness improves with mild pose jitter and color augmentation, but heavy spatial transforms that break ray geometry hurt. In practice, tuning group size and adding a few extra registers mitigate residual artifacts on wide baselines.

Budget-aware comparison. Matching wall-clock within 1 s at 128 views, a slightly smaller decoder preserves PSNR while lowering LPIPS, consistent with the accuracy–latency trend in Table 2.

6. Conclusion

We presented MVP, a dual-hierarchy multi-view transformer that scales inter-view context while compressing intra-view tokens. The design reduces attention dilu-

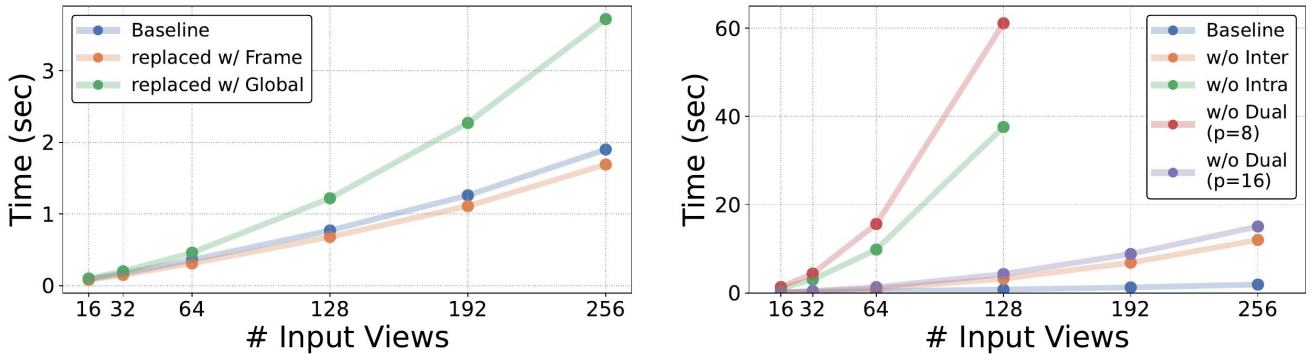

Figure 6. Ablations and inference time for MVP components.

Method	Views	Tanks & Temples			Mip-NeRF360		
		PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
Long-LRM [13]	32	18.59	0.614	0.366	21.08	0.484	0.445
iLRM [4]		18.58	0.631	0.385	21.09	0.495	0.466
Ours		**19.54**	**0.708**	**0.277**	**22.21**	**0.587**	**0.355**
Long-LRM	64	19.44	0.651	0.334	21.30	0.499	0.431
iLRM		19.82	0.692	0.318	21.60	0.522	0.444
Ours		**21.24**	**0.761**	**0.221**	**23.72**	**0.656**	**0.302**
Long-LRM	128	18.47	0.613	0.375	19.82	0.484	0.457
iLRM		19.22	0.696	0.319	21.32	0.551	0.424
Ours		**22.36**	**0.804**	**0.184**	**25.12**	**0.736**	**0.248**

Table 3. Quantitative comparisons on the Tanks&Temples and Mip-NeRF360 under a zero-shot setting. For the 128 view setting on Mip-NeRF360, the `stump` and `treehill` scenes are excluded due to their limited number of frames.

Method	4 views			8 views		
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑	LPIPS ↓
CLiFT [10]	30.13	0.916	0.095	29.68	0.910	0.092
iLRM [4]	30.37	0.923	0.095	28.90 (31.57)	0.920 (0.935)	0.102 (0.082)
Ours-coarse	30.56	0.925	0.087	31.78	0.939	0.077
Ours-fine	**32.12**	**0.941**	**0.077**	**33.40**	**0.952**	**0.066**

Table 4. Quantitative comparisons on the RE10K dataset. Baselines are evaluated using 4-view models, and values in parentheses $(\cdot)$ are from the models trained for that specific view setting.

tion, preserves global consistency, and achieves favorable accuracy–latency trade-offs up to 256 views. Experiments on DL3DV, Tanks&Temples, Mip-NeRF360, and RE10K demonstrate strong quality, speed, and zero-shot generalization. We hope MVP provides a practical blueprint for large-context, single-pass 3D Gaussian reconstruction.

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Method	32 views (Trained)				40 views (Unseen)				48 views (Unseen)			
	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	Time (s) ↓
Long-LRM [13]	23.97	0.778	0.267	0.84	24.18 (0.21)	0.787	0.260	1.05 (0.21)	24.30 (0.33)	0.797	0.252	1.38 (0.54)
iLRM [4]	24.30	0.803	0.257	0.53	24.54 (0.24)	0.811	0.248	0.76 (0.23)	24.78 (0.48)	0.820	0.240	1.04 (0.51)
Ours	**25.88**	**0.845**	**0.188**	**0.17**	**26.36** (0.48)	**0.855**	**0.178**	**0.22** (0.05)	**27.06** (1.18)	**0.870**	**0.164**	**0.26** (0.09)

Table 5. Longer context generalization. values in parentheses $(\cdot)$ denote the improvements over 32 views.

Method	Stage 1	Stage 2	Stage 3	Patch Size	Hidden Dim.	Feature Agg.	PSNR ↑	SSIM ↑	LPIPS ↓
Baseline	Frame(2)	Group(4)	Global(8)	(8, 16, 32)	(128, 256, 512)	o	22.79	0.733	0.235
w/o Feature Agg.	Frame(2)	Group(4)	Global(8)	(8, 16, 32)	(128, 256, 512)	x	21.58	0.646	0.340
w/o Group-attn.	Frame(2)	Frame(4)	Global(8)	(8, 16, 32)	(128, 256, 512)	o	22.53	0.720	0.247
	Frame(2)	Global(4)	Global(8)				22.94	0.739	0.235
w/o Inter-view Hierarchy	Global(2)	Global(4)	Global(8)	(8, 16, 32)	(128, 256, 512)	o	22.94	0.739	0.236
w/o Intra-view Hierarchy	Frame(2)	Group(4)	Global(8)	(8, 8, 8)	(512, 512, 512)	o	22.83	0.732	0.249
w/o Dual Hierarchy (p=8)	Global(2)	Global(4)	Global(8)	(8, 8, 8)	(512, 512, 512)	o	23.20	0.747	0.241
w/o Dual Hierarchy (p=16)	Global(2)	Global(4)	Global(8)	(16, 16, 16)	(512, 512, 512)	o	21.80	0.651	0.341
Reversed Hierarchy	Global(8)	Group(4)	Frame(2)	(32, 16, 8)	(512, 256, 128)	o	18.95	0.442	0.555

Table 6. Ablations. We train all variants for 100K iterations at a reduced resolution (256×256) on the DL3DV dataset.

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